

2.4.4 Explicit and Implicit Methods

All of the methods mentioned so far are said to be *explicit* discretizations, since producing the next approximation point on a trajectory does not require solving any implicit equations defined in terms of the previous one. The Verlet method is not implicit even though $\mathbf{Q}$ appears on the right in the second equation, since it is given in an explicit way in terms of $\mathbf{q}$ and $\mathbf{p}$. Implicitness adds another layer in both analysis and numerical implementation, which, however, in certain applications, is readily justified.

Example 2.6 The Backward Euler method,

$$\mathbf{z}_{n+1} = \mathbf{z}_n + h\mathbf{f}(\mathbf{z}_{n+1}),$$

is an example of an implicit method. The calculation of a timestep involves solving a system of equations of the form

$$\mathbf{g}(\mathbf{w}) \equiv \mathbf{w} - \mathbf{z}_n - h\mathbf{f}(\mathbf{w}) = 0.$$

The map $\mathcal{G}_t$ is defined implicitly by the equation

$$\mathcal{G}_h(\mathbf{z}) = \mathbf{z} + h\mathbf{f}(\mathcal{G}_h(\mathbf{z})).$$

An implicit method will typically result in a system of nonlinear equations of the form

$$\mathbf{g}(\mathbf{z}_{n+1}) = \boldsymbol{\tau}_n, \tag{2.25}$$

which will need to be solved at each timestep. The right hand side $\boldsymbol{\tau}_n$ is a vector that depends on the previous time-step $\mathbf{z}_n$, perhaps in a complicated way. We may assume the number of equations represented by (2.25) is equal to the dimension of the space where $\mathbf{z}$ is defined, so we have a square nonlinear system. Typically $\mathbf{g}$ will depend on the stepsize and coefficients of the method and will have the property that for h sufficiently small, the solution is uniquely defined and is continuously defined in terms of $\boldsymbol{\tau}_n$, that is the mapping $\mathbf{g}$ has a bounded and smooth inverse.

Solving the system may then proceed, from an initial guess $\mathbf{z}_{n+1}^{(0)}$ by use of Newton's method:

$$\mathbf{z}_{n+1}^{(k+1)} = \mathbf{z}_{n+1}^{(k)} - [\mathbf{J}^{(k)}]^{-1} (\mathbf{g}(\mathbf{z}_n^{(k)}) - \boldsymbol{\tau}_n),$$

where $\mathbf{J}^{(k)}$ is the Jacobian matrix of the mapping $\mathbf{g}$ evaluated at $\mathbf{z}_n^{(k)}$, or else an approximation of this Jacobian (assumed to be nonsingular due to the invertibility